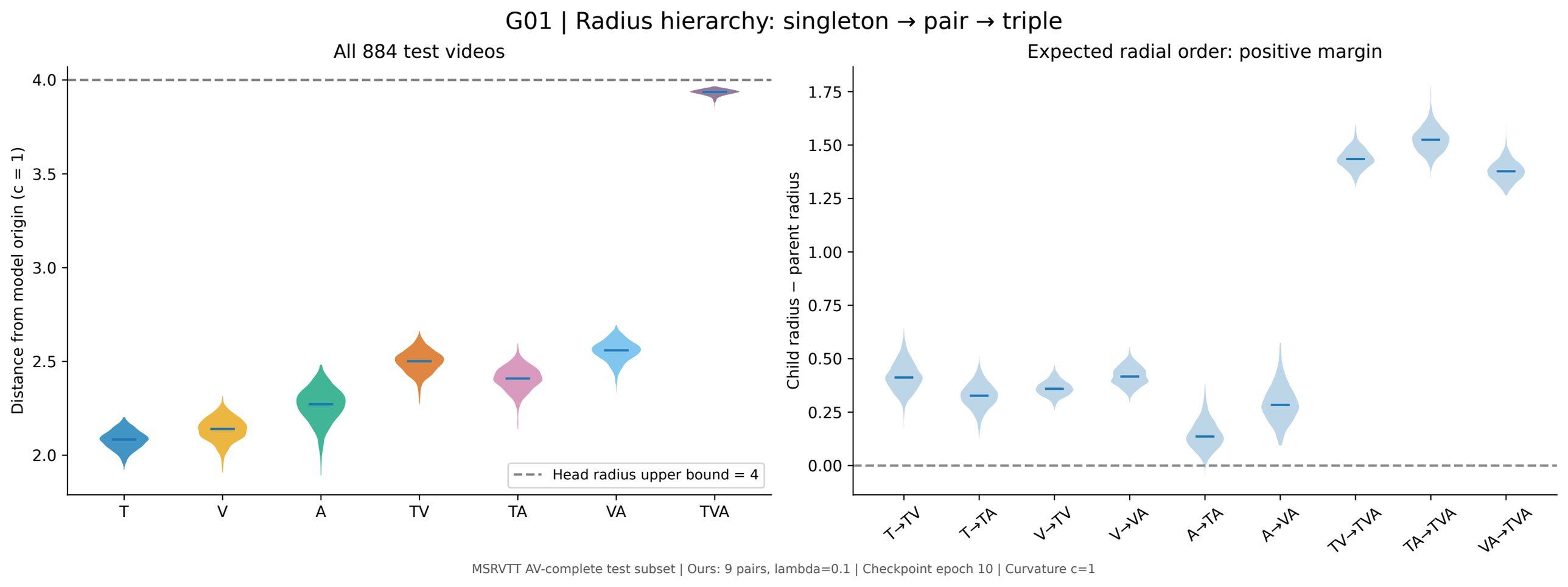
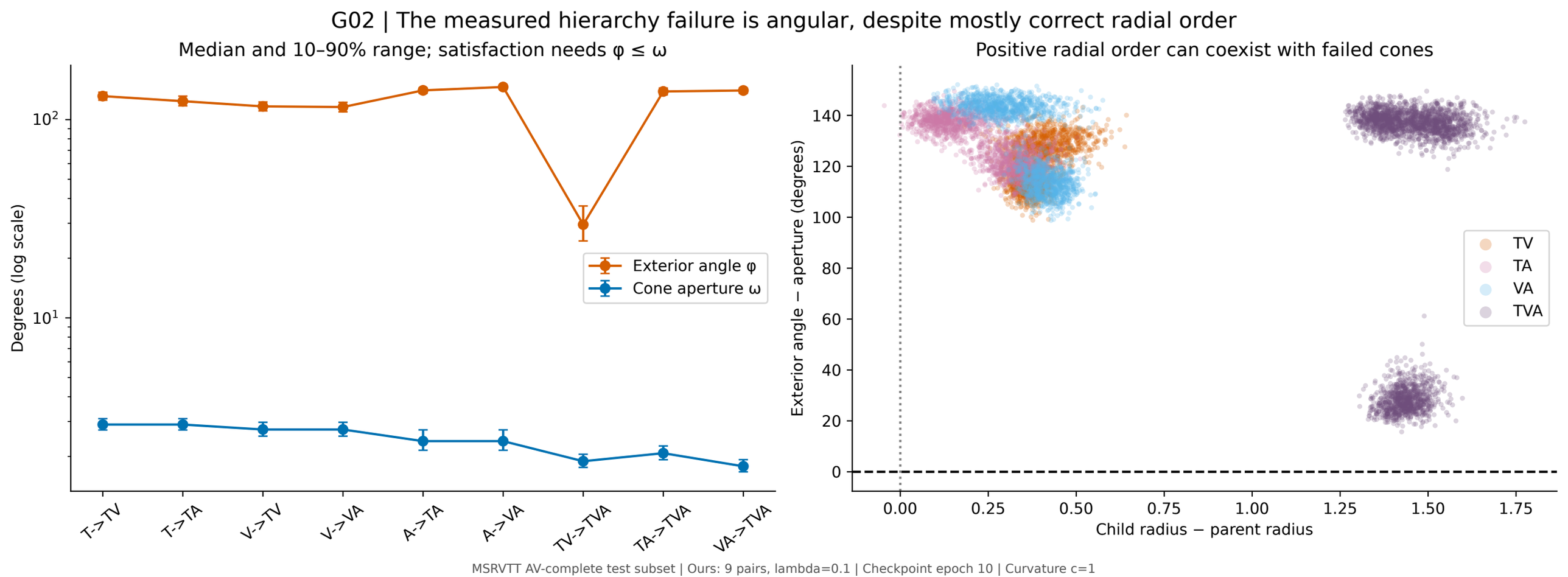
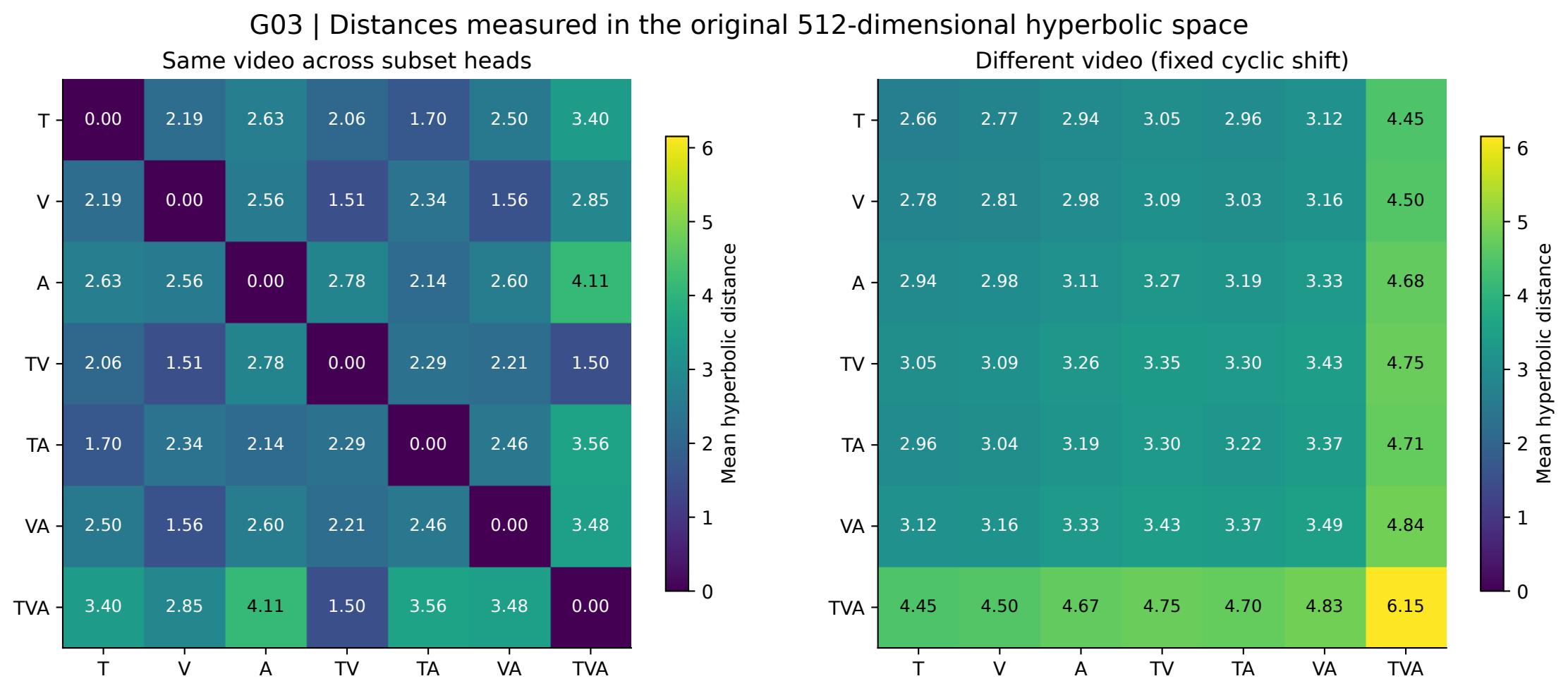
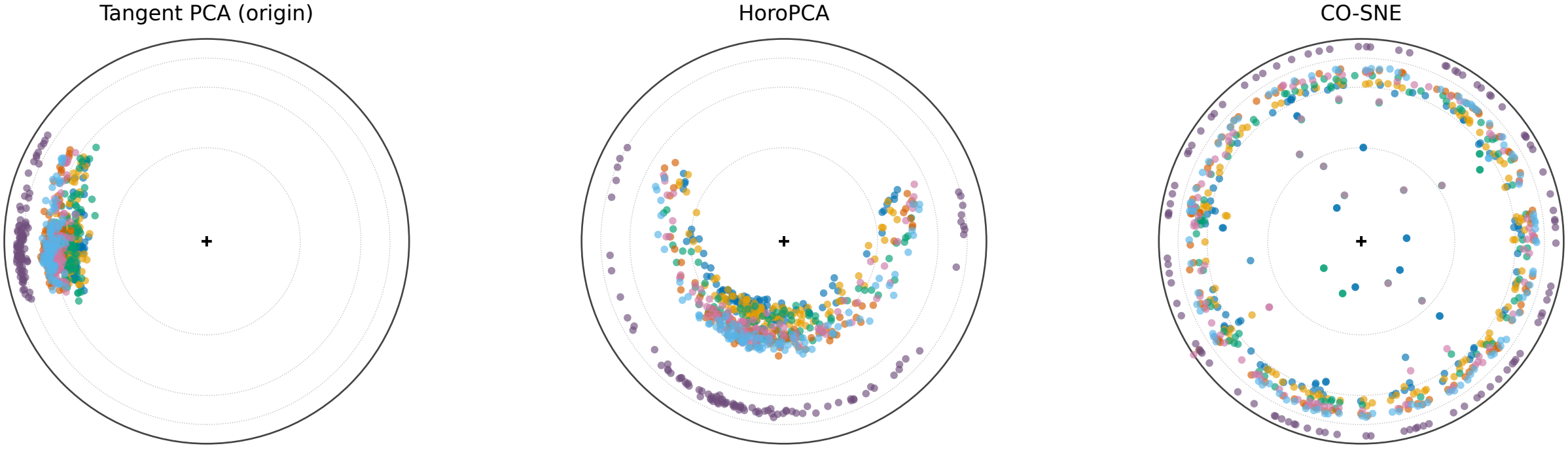

G01 | Radius hierarchy: singleton → pair → triple
All 884 test videos
Distance from model origin (c = 1)
4.0
3.5
3.0
2.5
2.0
Head radius upper bound = 4
T
V
A
TV
TA
VA
TVA
Expected radial order: positive margin
Child radius − parent radius
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
T→TV
T→TA
V→TV
V→VA
A→TA
A→VA
TV→TVA
TA→TVA
VA→TVA
MSRVTT AV-complete test subset | Ours: 9 pairs, lambda=0.1 | Checkpoint epoch 10 | Curvature c=1

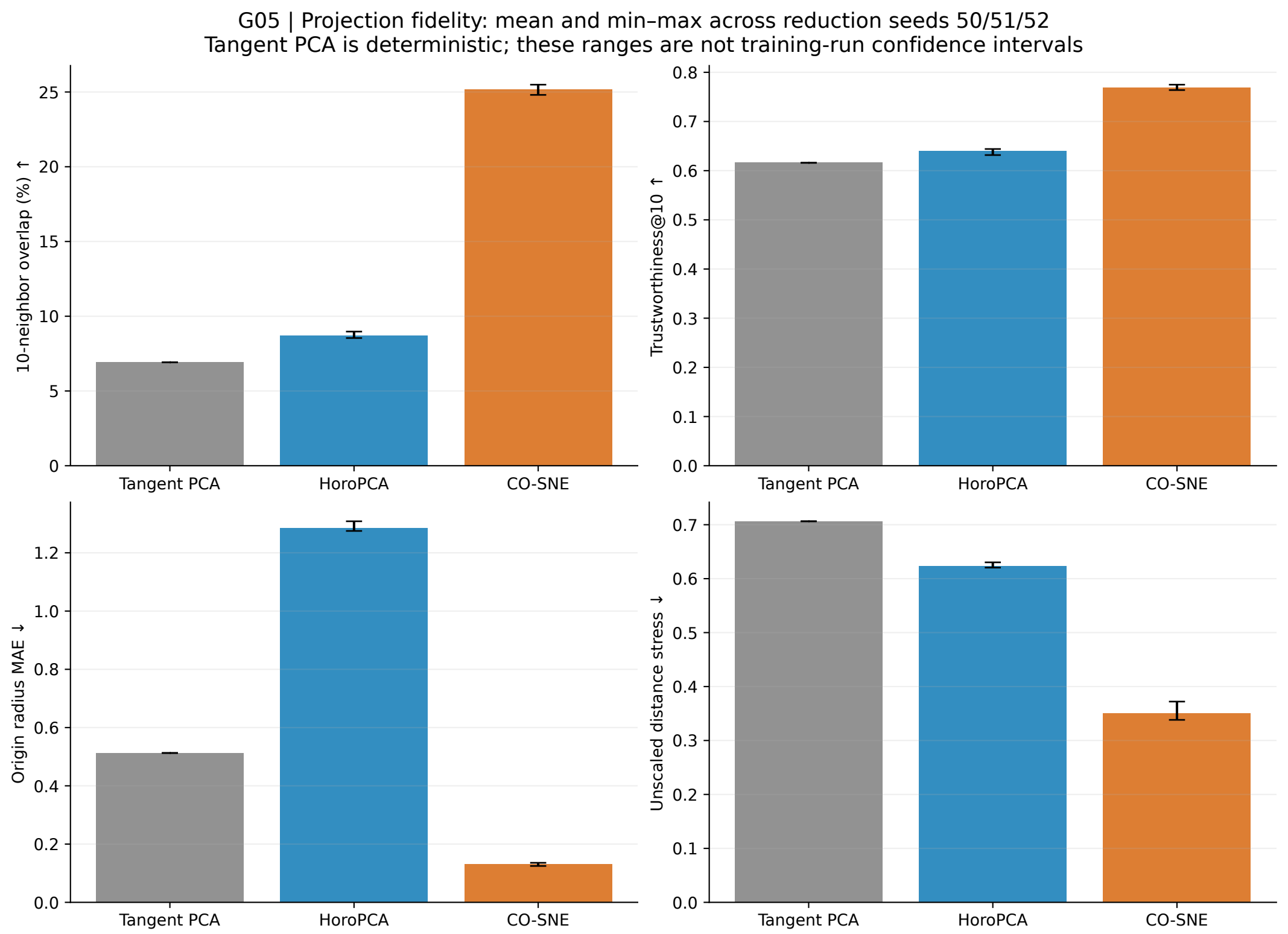

G02 | The measured hierarchy failure is angular, despite mostly correct radial order
Median and 10–90% range; satisfaction needs φ ≤ ω
Degrees (log scale)
10^2
10^1
Exterior angle φ
Cone aperture ω
T->TV
T->TA
V->TV
V->VA
A->TA
A->VA
TV->TVA
TA->TVA
VA->TVA
Positive radial order can coexist with failed cones
Exterior angle − aperture (degrees)
0
20
40
60
80
100
120
140
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
Child radius − parent radius
TV
TA
VA
TVA
MSRVTT AV-complete test subset | Ours: 9 pairs, lambda=0.1 | Checkpoint epoch 10 | Curvature c=1

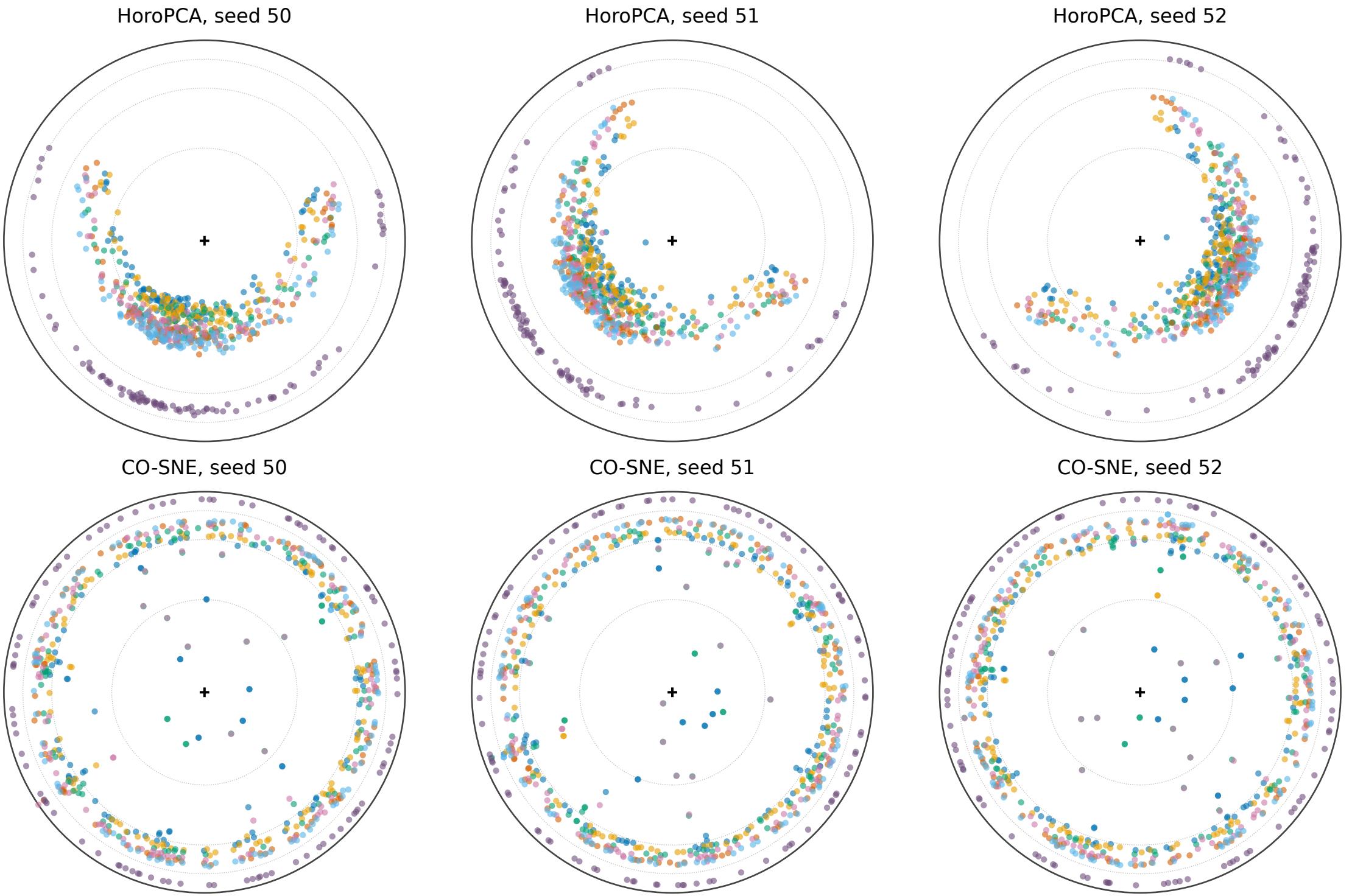

G03 | Distances measured in the original 512-dimensional hyperbolic space
Same video across subset heads
Different video (fixed cyclic shift)
T 0.00 2.19 2.63 2.06 1.70 2.50 3.40
V 2.19 0.00 2.56 1.51 2.34 1.56 2.85
A 2.63 2.56 0.00 2.78 2.14 2.60 4.11
TV 2.06 1.51 2.78 0.00 2.29 2.21 1.50
TA 1.70 2.34 2.14 2.29 0.00 2.46 3.56
VA 2.50 1.56 2.60 2.21 2.46 0.00 3.48
TVA 3.40 2.85 4.11 1.50 3.56 3.48 0.00
T V A TV TA VA TVA
Mean hyperbolic distance
0 1 2 3 4 5 6
T 2.66 2.77 2.94 3.05 2.96 3.12 4.45
V 2.78 2.81 2.98 3.09 3.03 3.16 4.50
A 2.94 2.98 3.11 3.27 3.19 3.33 4.68
TV 3.05 3.09 3.26 3.35 3.30 3.43 4.75
TA 2.96 3.04 3.19 3.30 3.22 3.37 4.71
VA 3.12 3.16 3.33 3.43 3.37 3.49 4.84
TVA 4.45 4.50 4.67 4.75 4.70 4.83 6.15
T V A TV TA VA TVA
Mean hyperbolic distance
0 1 2 3 4 5 6
MSRVTT AV-complete test subset | Ours: 9 pairs, lambda=0.1 | Checkpoint epoch 10 | Curvature c=1

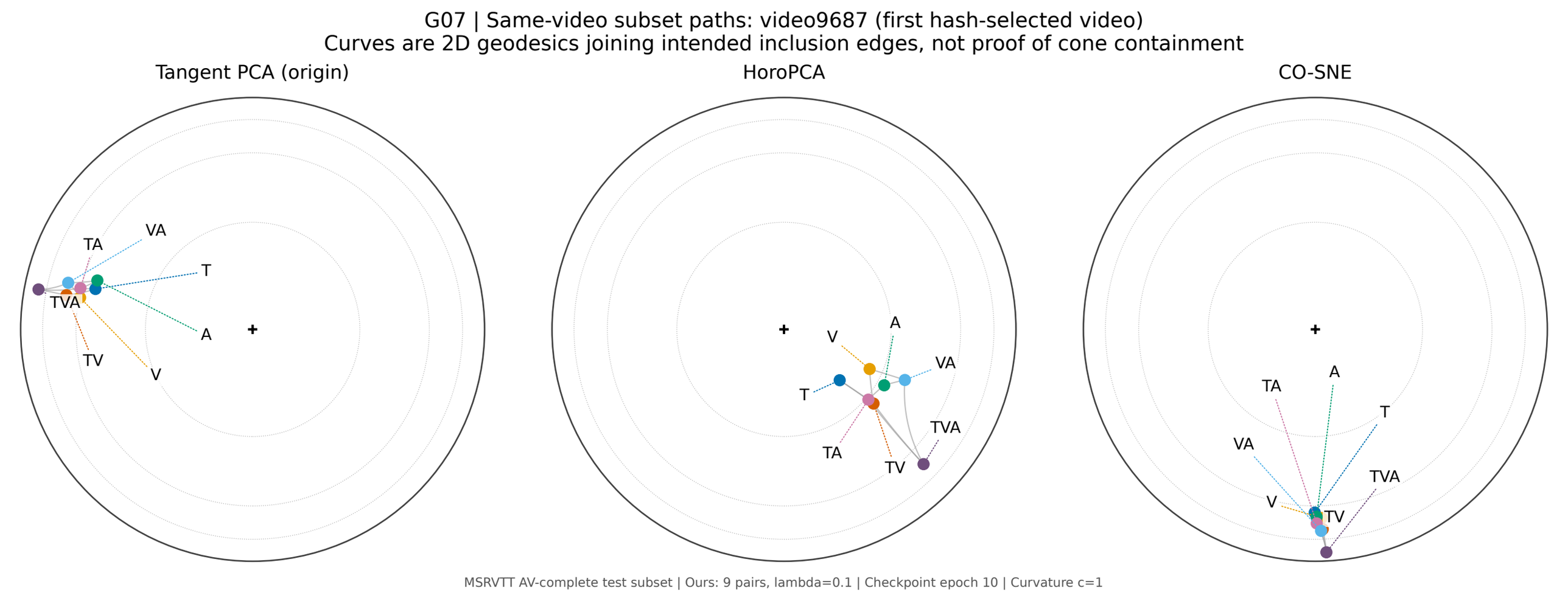
G04 | Same 96 videos × 7 heads; fixed display seed 50; colors indicate subset type
Unit Poincaré disk; dotted rings have hyperbolic radii 1, 2, 3

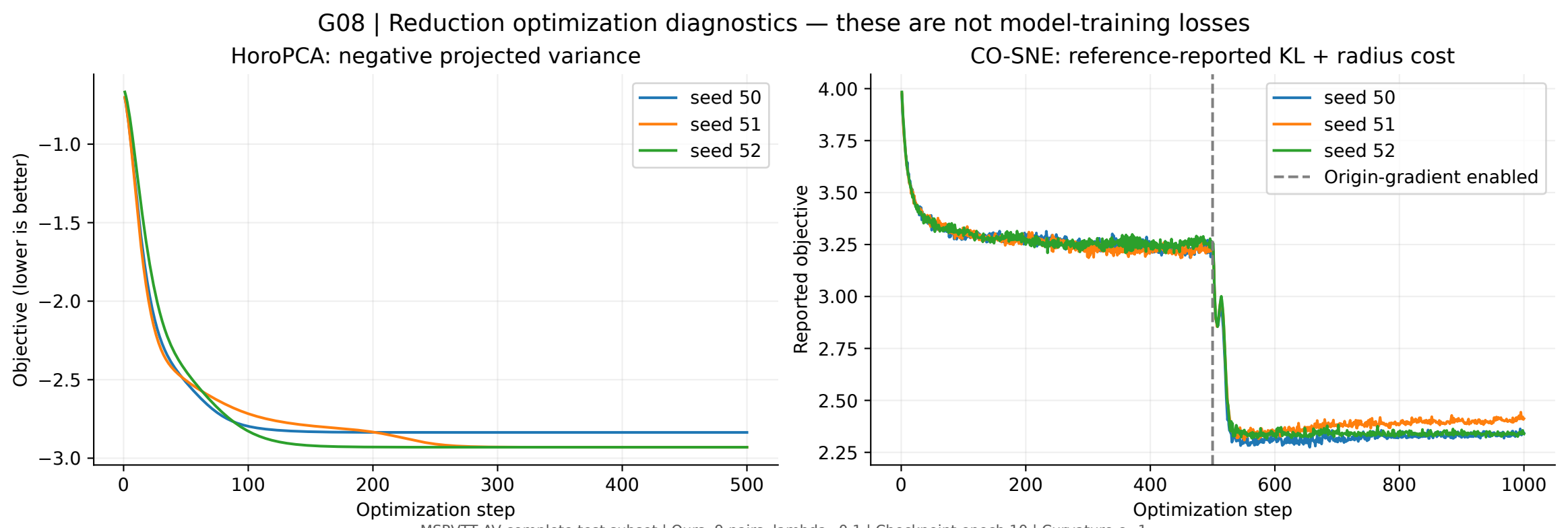


MSRVTT AV-complete test subset | Ours: 9 pairs, lambda=0.1 | Checkpoint epoch 10 | Curvature c=1


G05 | Projection fidelity: mean and min–max across reduction seeds 50/51/52
Tangent PCA is deterministic; these ranges are not training-run confidence intervals
10-neighbor overlap (%) ↑
Trustworthiness@10 ↑
Origin radius MAE ↓
Unscaled distance stress ↓
Tangent PCA
HoroPCA
CO-SNE
MSRVTT AV-complete test subset | Ours: 9 pairs, lambda=0.1 | Checkpoint epoch 10 | Curvature c=1


G06 | Seed sensitivity — orientation and apparent clusters can vary with reduction
HoroPCA, seed 50
HoroPCA, seed 51
HoroPCA, seed 52
CO-SNE, seed 50
CO-SNE, seed 51
CO-SNE, seed 52
MSRVTT AV-complete test subset | Ours: 9 pairs, lambda=0.1 | Checkpoint epoch 10 | Curvature c=1


G07 | Same-video subset paths: video9687 (first hash-selected video)
Curves are 2D geodesics joining intended inclusion edges, not proof of cone containment
Tangent PCA (origin)
HoroPCA
CO-SNE
VA
TA
T
TVA
A
TV
V
A
V
VA
T
TVA
TA
TV
A
TA
T
VA
TVA
V
TV
MSRVTT AV-complete test subset | Ours: 9 pairs, lambda=0.1 | Checkpoint epoch 10 | Curvature c=1


G08 | Reduction optimization diagnostics — these are not model-training losses
HoroPCA: negative projected variance
Objective (lower is better)
−1.0
−1.5
−2.0
−2.5
−3.0
0
100
200
300
400
500
Optimization step
seed 50
seed 51
seed 52
CO-SNE: reference-reported KL + radius cost
Reported objective
4.00
3.75
3.50
3.25
3.00
2.75
2.50
2.25
0
200
400
600
800
1000
Optimization step
seed 50
seed 51
seed 52
Origin-gradient enabled
MSRVTT AV-complete test subset | Ours: 9 pairs, lambda=0.1 | Checkpoint epoch 10 | Curvature c=1